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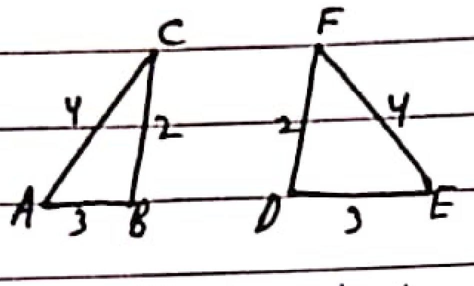
Circles.

→ **Rules of congurence :-**

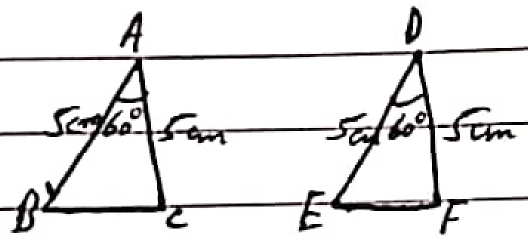
→ **Rule 1:-** SSS. (Side Side Side).



$AB = DE = 3cm$
$AC = EF = 4cm$
$BC = BF = 2cm$

∴ ΔABC is congurent to ΔDEF ($\Delta ABC \cong \Delta DEF$).

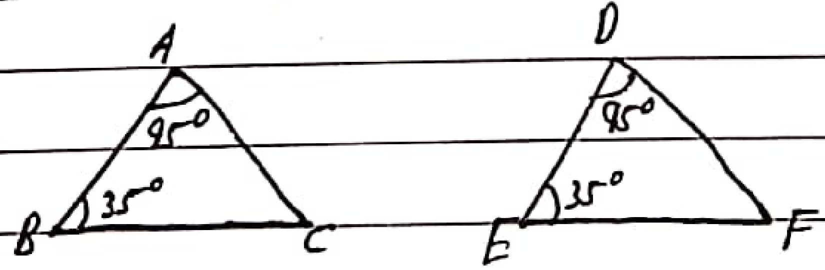
→ **Rule 2:-** SAS (Side Angle Side)



$\angle A = \angle D = 60°$
$AB = DE = 5cm$
$AC = DF = 5cm.$

∴ $ABC \cong \Delta DEF$

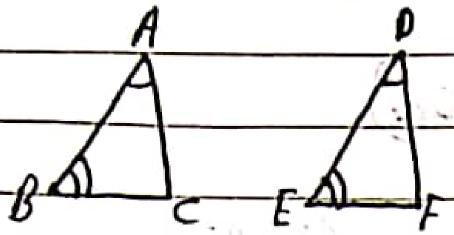
→ **Rule-3:-** ASA (Angle side Angle).



~~AB = BC~~ $\angle A = \angle D = 95°$
$\angle B = \angle E = 35°$
$AB = DE$

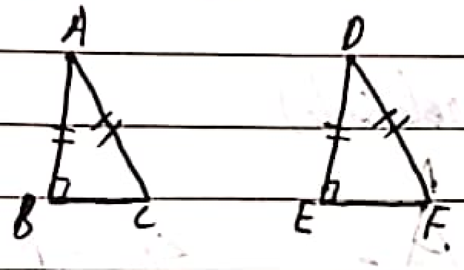
∴ $\Delta ABC \cong \Delta DEF$

→ <u>Rule 4:- AAS (Angle Angle side).</u>



$\angle A = \angle D$
$\angle B = \angle E$
$BC = EF$
$\Delta ABC \cong \Delta DEF$.

→ <u>RHS Rule:-</u> (Right Hypotneaus Side)

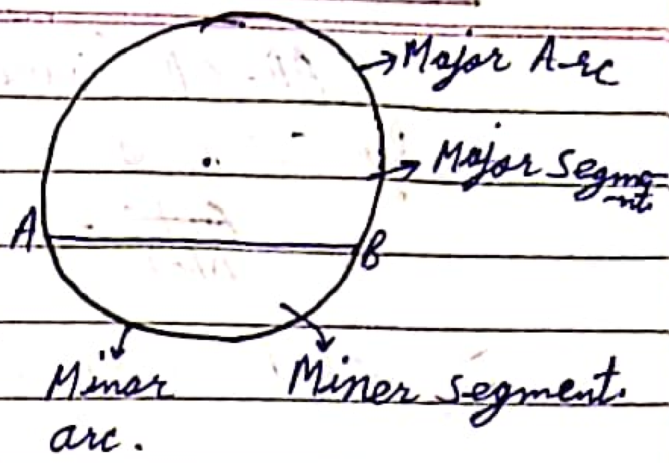


$AC = DF$
$AB = DE$

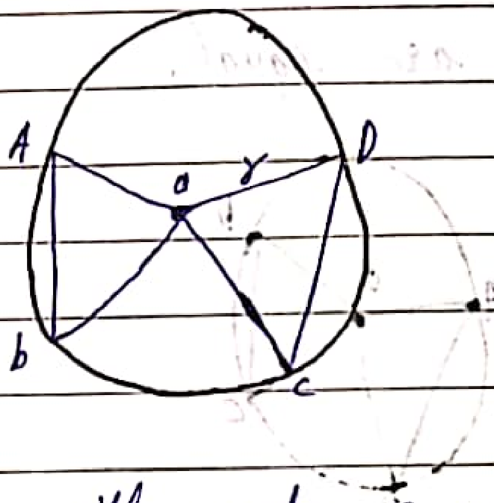
$\Delta ABC \cong \Delta DEF$.

⇒ <u>Circle</u>- Circle is the infinite set of points equidistant from a fixed point.

→ <u>Chord</u>- A line segment with its end points on the boundary of the circle.



- Theorm 9.1: Equal chords of a circle subtend equal angles at the centre.



- Given: Circle with centre O and radius r. i.e. C(O, r).
Chords AB and CD are such that:
AB = CD.

- To Prove: ∠AOB = ∠COD

- Proof: In ΔAOB and ΔCOD
AO = OD (Radii of same circle)
OB = OC (Radii of same circle)

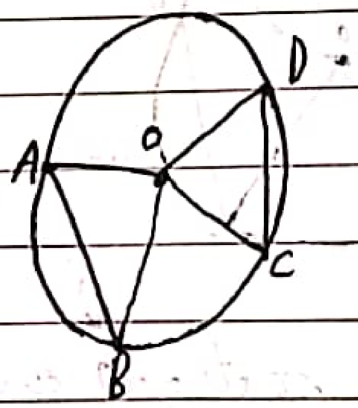
$AB = CD$ (Given)

$\therefore$ By SSS congurence

$\cancel{\Delta ABC}$ $\Delta AOB \cong \Delta COD$.

$\therefore$ $\angle AOB = \angle COD$ (C.P.C.T)

→ **Theorm: 9.2** - If the Angles subtended by the chords of a circle at the centre are equal, then the chords are equal.



→ Given: A circle with centre O, chords AB and CD, such that,

$\angle AOB = \cancel{\angle D} \angle COD$.

→ To Prove - $AB = CD$

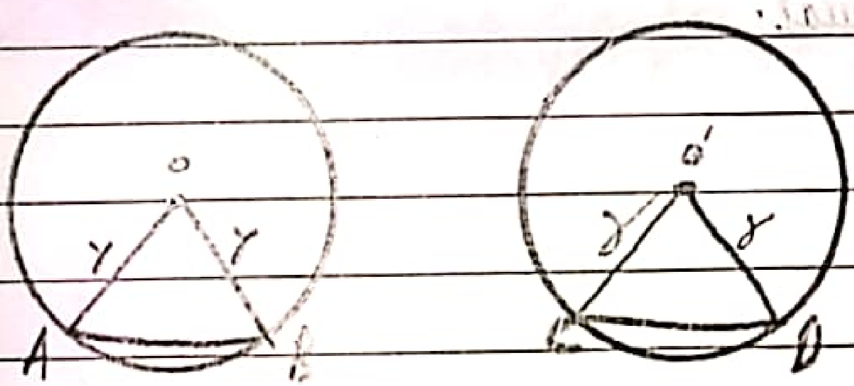
→ Proof - In ΔAOB and ΔCOD.

$AO = OD$ (Radii of the same circle).
$OB = OC$ (Radii of the same circle).
$\angle AOB = \angle COD$ (Given)
$\therefore$ By SAS congurence.
$\Delta AOB \cong \Delta COD$
$AB = CD$ (C.P.C.T)

EXERCISE = 8.1

Q1) Recall that two circles are congruent if they have the same radii...?

Sol.:-



Given:- Two Circles
$C_1(O, r)$ and
$C_2(O, r)$
- Chords AB of C_1 and CD of C_2 are, such that.
$AB = CD$.

→ **To prove** - $\angle AOB = \angle CO'D$

- **Proof** - In $\triangle AOB$ and $\triangle CO'D$

$AO = CO' = r$ (circles are congurent)

$BO = DO = r$ (circles are congurent)

$AB = CD$ (Given)

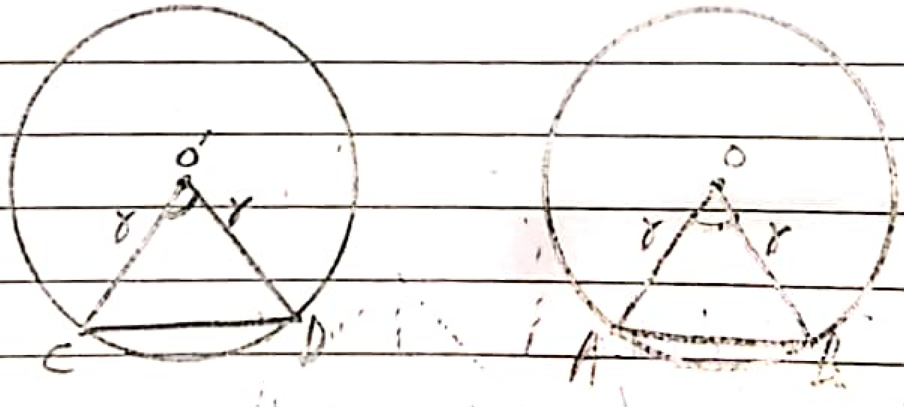
$\therefore$ By SSS congurence

$\triangle AOB \cong \triangle CO'D$.

$\angle AOB = \angle CO'D$ (C.P.C.T).

Q2) Prove that if chords of congurent circles subtend equal angles at their centres, then the chords are equal.

Sol:



- **Given** - Two circles.

Circle 1 (O', r)

Circle 2 (O, r)

$\therefore$ $\angle CO'D = \angle AOB$

- <u>To prove</u>. $CD = AB$

- <u>Proof</u>. In ΔAOB and $\Delta CO'D$

$\angle CO'D = \angle AOB$ (Given)

$CO' = AO = r$

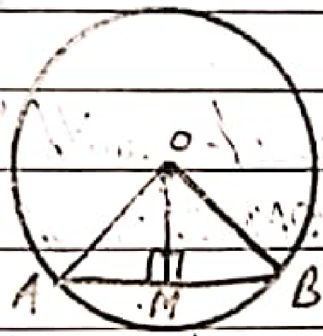
$BO = DO' = r$

$\Rightarrow$ By SAS Congurence

$\Delta AOB \cong \Delta CO'D$

$AB = CD$ (C.P.C.T)

$\Rightarrow$ <u>Theorm 9.3</u> - <u>The Perpendicular from the centre of a circle to a chord bisects the chord.</u>



- <u>Given</u>:- A circle with centre O, Chord AB and $OM \perp AB$.

- <u>To prove</u>- $AM = MB$

- Const. Join OA and OB
- Proof. In ΔAMO and ΔBMO

$OA = OB$ (Radii of same circle)

$OM = OM$ (Common)

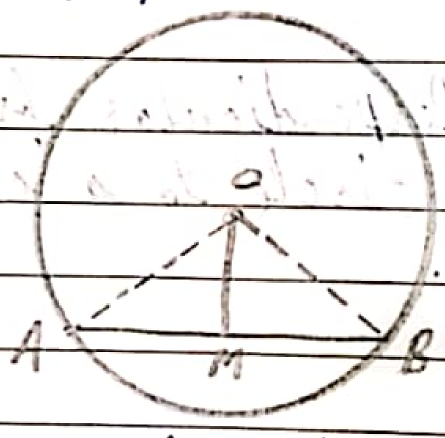
$\angle AMO = \angle BMO$ (Each 90°)

$\Rightarrow$ By SSA Congurence

$\Delta AMO \cong \Delta BMO$

$AM = BM$ (C.P.C.T)

$\Rightarrow$ Theorm (9.4). A line drawn from the centre of the circle to bisect the chord is perpendicular to the chord.



- Given. A circle with centre O, AB is chord and line OM through centre 'O', such that AM=MB
- To prove. $OM \perp AB$
- Const. Join AO and BO.

- Proof- In ΔAMO and ΔBMO

$AO = BO$ (Radii of same circle)

$OM = OM$ (Common)

$AM = MB$ (Given)

- By SSS congurence

$\Delta AMO \cong \cancel{AB}\ \Delta BMO$

$\angle AMO = \angle BMO = x$.

- Now, $\angle AMO + \angle BMO = 180^\circ$

$x + x = 180^\circ$

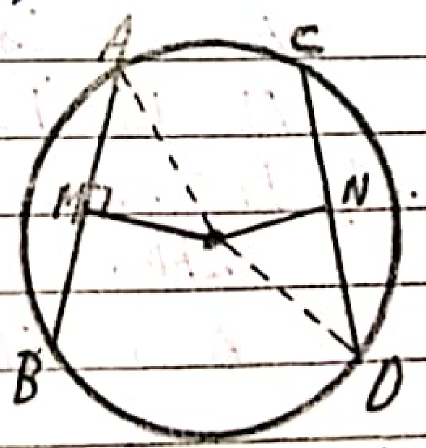
$2x = 180^\circ$

$x = \frac{180^\circ}{2} = 90^\circ$

$\therefore$ $\angle AMO = \angle BMO = x = 90^\circ$

$\Rightarrow$ $OM \perp AB$.

$\Rightarrow$ Theorm(9.5) Equal chords of circle (or econgurent circle) are equidistant from the centre.



- Given- A circle with centre 'O'. AB and CD are

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two chords of the same circle.

$AB = CD$

$OM \perp AB$

$ON \perp CD$.

- To prove- $OM = ON$
- Const- Join AO and OB.
- Proof- Since $OM \perp AB \Rightarrow AM = MB$ ---(1)

$OM \perp CD \Rightarrow CN = ND$ ---(2)

$AB = CD$ (Given)

- $AM + MB = CN + ND$

$AM + AM = ND + ND$

$\cancel{2}AM = \cancel{2}ND$

$\boxed{AM = ND}$ --- (3)

- In ΔAMO and ΔDNO

$AO = DO$ (Equal Radii)

$AM = ND$ (Proved).

$\therefore$ By RHS congurence

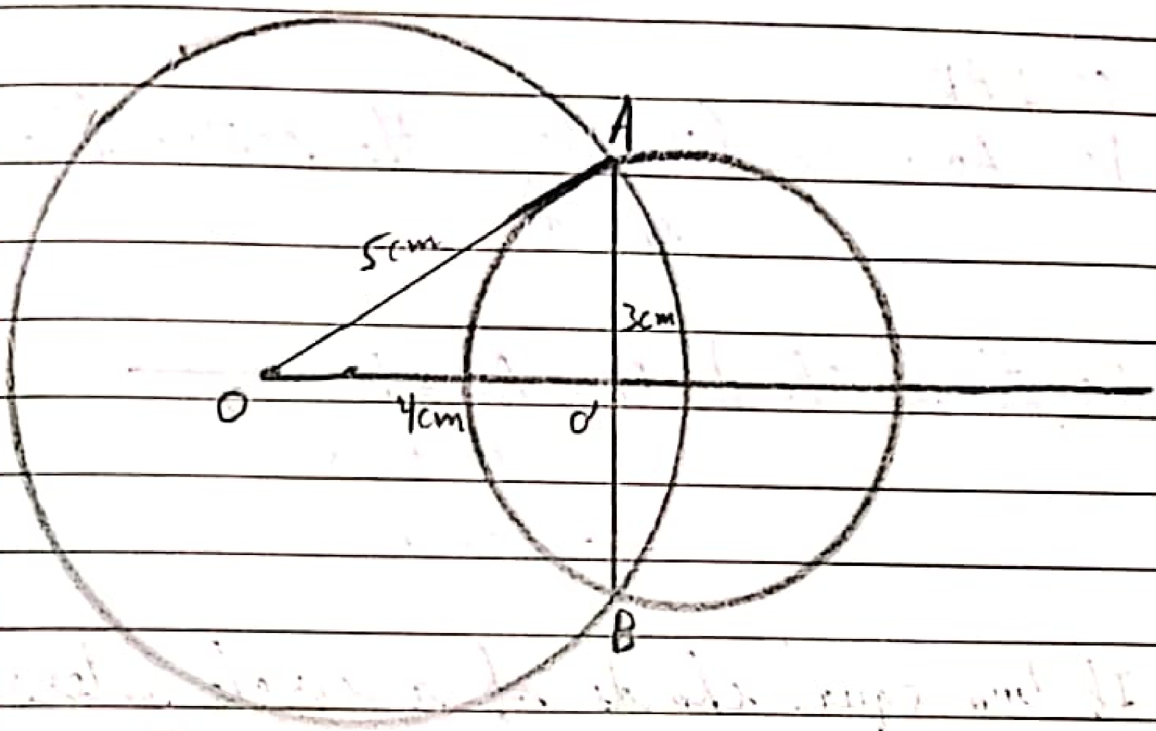
$\Delta AMO \cong \Delta DNO$

$\Rightarrow OM = ON$ (C.P.C.T).

Exercise = 8.2

Q1:- Two circles of radii 5cm and 3cm intersect at two points and the distance between ~~their~~ their centres is 4cm. Find the length of common chord.

Sol^n



- Given: Two circles with centre O and O', Radii 5cm and 30cm. Such that,

  $OO' = 4cm$

- Const - Join OA.

- In $\Delta OO'A$

$$(OA)^2 = 5^2 = 25 \quad ---- (1)$$
$$(OO')^2 = O'A = 4^2 + 3^2 = 16 + 9$$
$$= 25 \quad ---- (2)$$

- From (1) and (2), we have: $OA^2 = (OO')^2 + O'A^2$

i.e, $H^2 = B^2 + P^2$

$\Rightarrow$ $\Delta OO'A$ is a right angled Δ.

$\Rightarrow$ $OO' \perp AB$

$\Rightarrow$ $BO' = AO' = 3cm$ ($\because \perp$ from the centre bisects the chord).

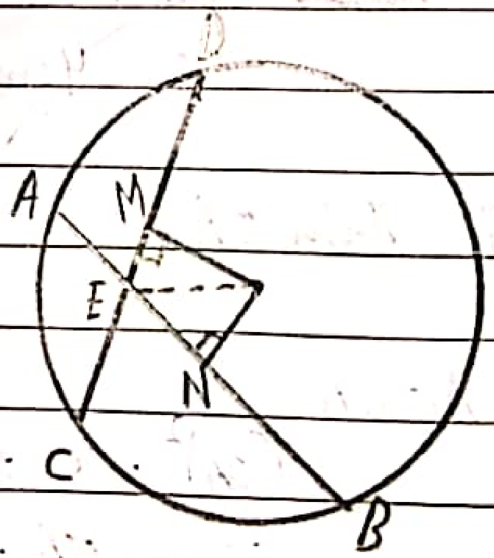
$\therefore$ Length of Chord $AB = AO' + BO'$

$= 3cm + 3cm$

$= 6cm.$

Q2~ If two equal chords of a circle intersect within the circle, ---?

Sol)



Given- A circle with centre 'O' chords AB and

CD intersect at point E. and $AB = CD$.

- To Prove- $ED = EB$
$CE = AE$.

- Const- 1) Draw $ON \perp AB$, $OM \perp CD$
2) Join EO.

- Proof- In ΔEMO and ΔENO
$\angle M = \angle N$ (Each 90°)
$OE = OE$ (Common)
$OM = ON$ (Chords Equidistant)
- By RHS congruence.
$\Delta EMO \cong \Delta ENO$.
$\Rightarrow EN = EM$ (C.P.C.T) --- ①

- Since $AN = \frac{1}{2}AB$ --- ② ($\because$ N is mid-point of AB)

and $CM = \frac{1}{2}CD$.

$\Rightarrow AB = 2AN$ --- ③
$\Rightarrow CD = 2CM$ --- ④

- Since $AB = CD$ (Given)
$\cancel{2}AN = \cancel{2}CM$ (by 3, 4)
$\Rightarrow AN = CM \Rightarrow \boxed{CM = AN}$ --- ⑤

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- subtracting (1) from (5)

- $CM - EN = AN - EN$

$CE = AE$

$\Rightarrow \boxed{EC = EA}$ ---- (6)

- Again $CD = AB$ --- (7) (Given)

$\Rightarrow$ subtracting (6) from (7)

$CD - EC = AB - EA$

$\boxed{ED = EB}$.